

Shot Clouds: Predicting Where an NBA Player Shoots in a Game

Aaron John Danielson
The University of Texas at Austin
aaron.danielson@austin.utexas.edu

Introduction

Scouting and lineup simulations need to know where a player will shoot in a game. The season shot chart [1] averages hundreds of attempts, while one game holds ten to thirty, so a single-game forecast is a distribution over small sets of shots, a *shot cloud*. For rookies and players in new roles, own history is thin, so the forecast must borrow from similar players. We ask whether learning which individual past attempts matter for a given game yields more realistic shot clouds.

Methods

Using public NBA shot data (2014–2024), we train through June 2023 and evaluate on 2023–24. Each player-game is a marked point process of shot count, times and locations. Our central contribution is shot-level attention. Adaptive Collaborative KDE (AC-KDE) uses context-dependent attention to weight past shots from the player and from similar players, forming the predicted location density. A gate increases the weight on the player’s own shots as his history grows, and the weights reveal which past attempts shape each forecast (Figure 1). Predictions are conditional: they take the player’s observed minutes and shot count and, at each attempt, the game clock and his earlier observed shots that game; all other inputs are known before tipoff. We score simulated clouds on six distances against a noise floor from resampling the observed cloud.

Results

On the 1,892 high-volume player-games every method scores (mean 22 shots), AC-KDE produces the most realistic clouds on all six measures, ahead of a position-only chart, the player’s own shot chart and a mixture density network (every paired 95% interval excludes zero; Figure 2). Mean shot-distance error is 3.00 feet for the position chart, 2.65 for the player’s chart and 2.49 for AC-KDE. The network has lower per-shot negative log-likelihood (6.06 versus 6.15 nats) yet produces less realistic clouds: likelihood alone would have picked the wrong model. Replacing learned attention with equal weights on the same shots worsens all six measures (shot-distance error 2.49 to 2.58 feet). The weight on a player’s own shots rises from 0.64 (under 92 prior attempts) to 0.92 (over 423).

Conclusion

The model learns which real shots to remember and how much weight to give each. Shot clouds give teams simulation-ready predictions for players with varying amounts of history, judged one game at a time.

References

[1] Goldsberry, K. (2012). CourtVision. *MIT Sloan Sports Analytics Conference*.

Code: <https://github.com/aaronjdanielson/shot-cloud-generation>

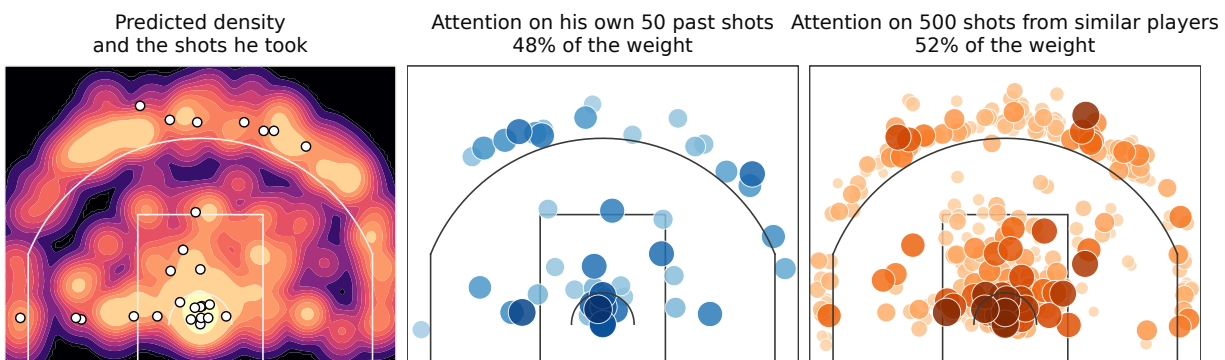


Figure 1: AC-KDE forecast for one rookie game (Victor Wembanyama, 26 shots). Left: predicted density, observed shots in white. Middle and right: each dot is a past shot; larger, darker dots carry more attention.

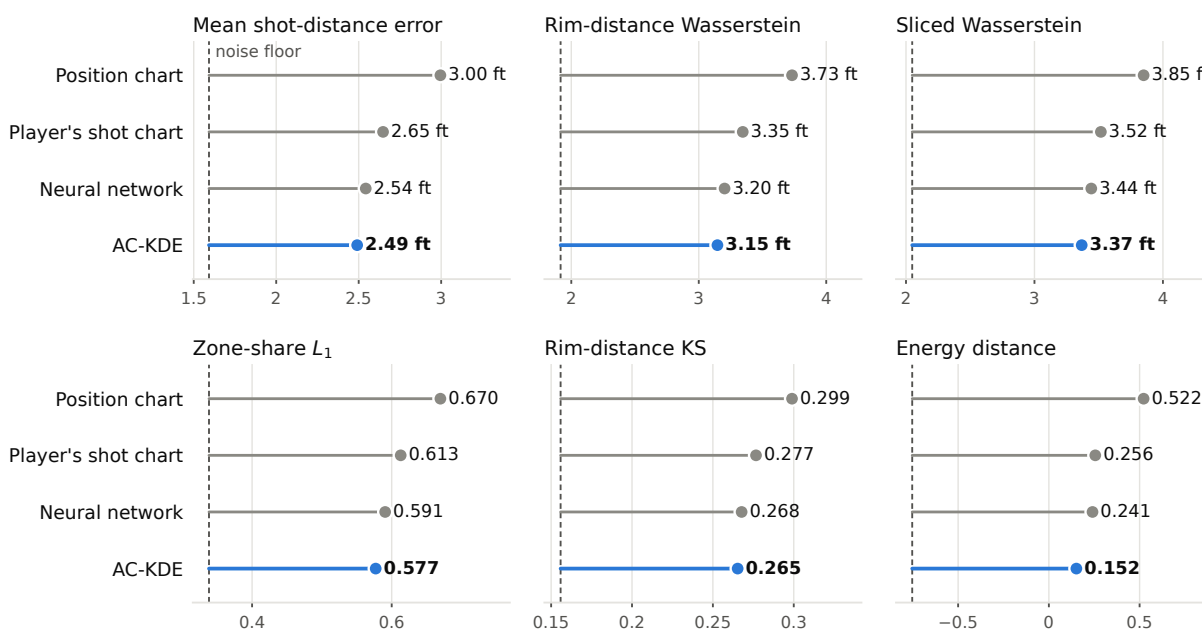


Figure 2: Shot-cloud error on the same 1,892 player-games from 2023–24 (lower is better; dashes: noise floor). Position chart: his position group's shot density.